Hand in 2

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Sorry for the miniuscule text, the tables overflowed when I tried to make it bigger. Also, the links to propositions and such do not provide much contrast to the cell colors, but they should be clickable in most PDF readers.

Exercise 1(d–j): Properties of particular spaces

	I_o^2	$\mathbb{R}_\ell$	$\mathbb{R}_\ell^2$	$\mathbb{R}^\omega$ product	$\mathbb{R}^\omega$ uniform	$\mathbb{R}^\omega$ box	$\mathbb{R}^{[0,1]}$ product
Connected	Example 6, §24	$[0,\infty)$	$[0,\infty)\times\mathbb{R}_\ell$	Exercise 10, §23	See proposition 1.1	Example 6, §23	Exercise 10, §23
Path connected	Example 6, §24	↑	↑	Interpolation	Implied by above	↑	
Locally connected	Exercise 3, §25	$[y,\infty)$	$\prod_i[y_i,\infty)$	↓	See proposition 1.2	*Implied by below*	
Locally path connected	Exercise 3, §25	↑	↑	Interpolation	*See proposition 1.2*	*See proposition 1.7*	
Compact	Thm 27.1	$\{[n,n+1)\}_{n\in\mathbb{Z}}$	←	π_i onto $\mathbb{R}$	π_i onto $\mathbb{R}$	π_i onto $\mathbb{R}$	π_i onto $\mathbb{R}$
Locally compact Hausdorff	Example 3, §29	*See proposition 1.5*	*See corollary 1.1*	Example 2, §29	Exercise 4, §29	*Example 2 §29*	Example 2, §29 (adapt)
Hausdorff	Order topology	$x\neq y\to[x,t),[t,y+1)$	Thm 31.2(a)	Thm 31.2(a)	Metrizable	Refinement of product	Thm 31.2(a)
Regular	↓	↓	↓	↓	↓		Thm 31.2(b)
Completely regular	↓	↓	Example 1, §33	↓	↓		Thm 33.2
Normal	Thm 32.3	Example 2, §31	Example 3, §31	Thm 32.2	Thm 32.2	N/A	Example 1, §32
Metrizable	See proposition 1.3	*see Exercise 5(a), §30*	←	Thm 20.5	Uniform metric		
Second-countable	See proposition 1.4	*See proposition 1.6*	←	Thm 30.2	Example 2, §30	Example 2, §30 (adapt)	$\{0,1\}^{[0,1]}$?

Exercises 2–6: General preservation properties

	Metric space	Compact Hausdorff	Arbitrary subspace	Open subspace	Closed subspace	Finite products	Countable products	Arbitrary products	Continuous image
Connected	$\{0,1\}_{\text{disc}}$	$\{0,1\}_{\text{disc}}$	$X\cup Y$	$X\cup Y$	$\{0,1\}\subset[0,1]$	Thm 23.6	Exercise 10, §23	Exercise 10, §23	Thm 23.5
Path connected	↑	↑	*counterexample*	*counterexample*	*counterexample*	Coordinate paths			$f\circ p$
Locally connected									
Locally path connected									
Compact	$\mathbb{R}$	By hypothesis	$(0,1)\subset[0,1]$	$(0,1)\subset[0,1]$	Thm 26.2	Tychonoff	Tychonoff	Tychonoff	Thm 26.5
Locally compact Hausdorff									
Hausdorff	Metric	By hypothesis	Thm 31.2(a)	Thm 31.2(a)	Thm 31.2(a)	Thm 31.2(a)	Thm 31.2(a)	Thm 31.2(a)	*counterexample*
Regular	↓	↓	Thm 31.2(b)	Thm 31.2(b)	Thm 31.2(b)	Thm 31.2(b)	Thm 31.2(b)	Thm 31.2(b)	
Completely regular	↓	↓	Thm 33.2	Thm 33.2	Thm 33.2	Thm 33.2	Thm 33.2	Thm 33.2	
Normal	Thm 32.2	Thm 32.3						Example 1, §32	
Second-countable	$\mathbb{R}^\omega$ uniform		Thm 30.2	Thm 30.2	Thm 30.2			$\mathbb{R}^{[0,1]}$	
Metrizable	By definition		Restriction of metric	Restriction of metric	Restriction of metric				

1 Proofs

Proposition 1.1. *$\mathbb{R}^\omega$ with the uniform topology is **not connected**.*

Proof. The countably infinite product of the real line $\mathbb{R}^\omega$ with the uniform topology is the space induced by the uniform metric

$$\bar{d}(x, y) = \sup_{n \in \mathbb{N}} \min(|x_n - y_n|, 1)$$

where $x = (x_n)$ and $y = (y_n)$ are sequences of real numbers.

We claim this space is **not connected**.

Let B be the set of all bounded sequences, and let U be the set of all unbounded sequences,

1. $B = \{x \in \mathbb{R}^\omega \mid \exists M > 0 \text{ s.t. } |x_n| \leq M \ \forall n \in \mathbb{N}\}$
2. $U = \mathbb{R}^\omega \setminus B$.

Clearly, these are nonempty since

$$(0, 0, 0, \ldots) \in B$$

while

$$(1, 2, 3, \ldots) \in U.$$

B is open.

Let $x = (x_n) \in B$. Then there exists some $M_x > 0$ such that $|x_n| \leq M_x$ for all $n \in \mathbb{N}$.

Consider the ball of radius 1 around x

$$B_{\bar{d}}(x, 1) = \{y \in \mathbb{R}^\omega \mid \bar{d}(x, y) < 1\}.$$

If $y \in B_{\bar{d}}(x, 1)$, then

$$\sup_{n \in \mathbb{N}} \min(|x_n - y_n|, 1) < 1$$

implying that for every n, $|x_n - y_n| < 1$ for all n. Using the triangle inequality we can attain the following bound:

$$|y_n| \leq |y_n - x_n| + |x_n| < 1 + M_x.$$

Thus the sequence y is bounded, meaning

$$B_{\bar{d}}(x, 1) \subseteq B.$$

Since every point of B has an open neighborhood completely contained in B, B is open.

U is open.

Let (x_n) be undounded. Consider the same open ball $B_{\overline{d}}(x, 1)$. If $y \in B_{\overline{d}}(x, 1)$ then

$$|y_n| \geq |x_n| - |x_n - y_n| > |x_n| - 1$$

so y is unbounded since $|x_n|$ can be arbitrarily large.

Therefore, $B_{\overline{d}}(x, 1) \subseteq U$ is open.

Then

$$\mathbb{R}^\omega = B \cup U.$$

■

Proposition 1.2. *$\mathbb{R}^\omega$ with the uniform topology is **locally connected**. In fact, it is **locally path connected**.*

Proof. We claim the space is **locally path connected**.

For any point $x = (x_n) \in \mathbb{R}^\omega$ and any radius $\varepsilon > 0$ the open ε-ball centered at x is

$$B_{\overline{d}}(x, \varepsilon) = \{y \in \mathbb{R}^\omega \mid \overline{d}(x, y) < \varepsilon\}.$$

If we choose $\varepsilon < 1$, the uniform metric forces every coordinate to be within ε of x_n. Therefore, the open ε-ball is

$$B_{\overline{d}}(x, \varepsilon) = \prod_{n \in \mathbb{N}} (x_n - \varepsilon, x_n + \varepsilon).$$

Take any point $y \in B_{\overline{d}}(x, \varepsilon)$. Construct a linear path $f : [0, 1] \to B_{\overline{d}}(x, \varepsilon)$ from the center point x to y by setting

$$f(t) = (1 - t)x + ty.$$

Since each interval $(x_n - \varepsilon, x_n + \varepsilon)$ is convex, every point remains inside the ball for all $t \in [0, 1]$.

Since every point has a neighborhood basis consisting of these path-connected open balls, the space is locally path-connected which makes it locally connected.

■

Proposition 1.3. *The ordered square I_o^2 is **not metrizable**.*

Proof sketch. We invoke the following theorem:

Theorem. *If a topological space is **compact** and **metrizable**, then it must be separable.*

So it will suffice to elucidate that the space is compact and not separable.

The Heine-Borel theorem for ordered spaces says that a linearly ordered topological space is compact if and only if it has a least element, a greatest element, and satisfies the least upper bound properrty.

The ordered square has minimum $(0,0)$, maximum $(1,1)$. Because the standard unit interval has the least upper bound property, the lexicographic order preserves this completeness. Thus I_o^2 is compact.

The space is not separable. We will find an uncountable collection of open sets that are completely isolated from each other. For every real $x \in (0,1)$ consider the vertical segment at x, excluding endpoints,

$$V_x = \{x\} \times (0,1).$$

In the dictionary order, V_x is the open interval $((x,0),(x,1))$. Because it is an open interval, V_x is an open set in the dictionary order topology. Notice that if $x \neq y$ the vertical segments V_x and V_y are disjoint,

$$V_x \cap V_y = \emptyset.$$

Since there are uncountably many choices of $x \in (0,1)$ we have an uncountable collection of pairwise disjoint non-empty open sets.

Suppose then that some dense $D \subseteq I_o^2$ exists. By definition the set must intersect all non-empty open sets, but there are uncountably many so this cannot be the case.

■

Proposition 1.4. *The ordered square I_o^2 is **not second-countable**.*

Proof sketch. As we saw in the proof of proposition 1.3, we can construct an uncountable collection of pairwise disjoint, non-empty open sets. By definition, each of these sets must contain at least one basic open set from the basis.

Since the vertical intervals are disjoint from one another, a basic open set in one cannot be shared by any other, meaning we must have uncountably many basis elements. ■

Proposition 1.5. *The Sorgenfrey line $\mathbb{R}_\ell$ is **not locally compact Hausdorff**.*

Proof by some guy on mathstackexchange said. In this Math stackexchange post there is given a proof that the Sorgenfrey line is not locally compact, and thus not locally compact Hausdorff. ■

Corollary 1.1. *The Sorgenfrey plane $\mathbb{R}_\ell^2$ is **not locally compact Hausdorff**.*

Proof sketch. Suppose it was. Considert the standard projection map onto coordinate one,

$$\pi_1 : \mathbb{R}_\ell \times \mathbb{R}_\ell \to \mathbb{R}_\ell$$

where

$$\pi_1(x, y) = x.$$

This map is continuous and open. The image of a locally compact space under a continuous open map is also locally compact, meaning $\mathbb{R}_\ell$ is locally compact, contradiction proposition 1.5.

Thus $\mathbb{R}_\ell^2$ is not locally compact and thus not locally compact Hausdorff. ■

Proposition 1.6. *The Sorgenfrey line $\mathbb{R}_\ell$ is **not second countable**.*

Proof. Suppose it is second-countable with basis

$$\mathscr{B} = \{B_1, B_2, \ldots\}.$$

For every $x \in \mathbb{R}$, $[x, x+1)$ is open. Since $\mathscr{B}$ is a basis there must exist, for every real $x \in \mathbb{R}$, an element $B_{n_x} \in \mathscr{B}$ such that

$$x \in B_{n_x} \subseteq [x, x+1].$$

Notice that x must be the absolute least element in $[x, x+1)$. Since $B_{n_x} \subseteq [x, x+1]$ and $x \in B_{n_x}$, x must be the absolute least element of B_{n_x}.

If we have two distinct $x, y \in \mathbb{R}$, then they cannot share the same basis element. If $B_{n_x} = B_{n_y}$ then these basis elements have two distinct minimum values, which is impossible.

Thus we can construct an injection $x \mapsto B_{n_x}$, meaning

$$\aleph_1 = |\mathbb{R}_\ell| \leq |\mathscr{B}|$$

so $\mathscr{B}$ is uncountable. ■

Proposition 1.7. *$\mathbb{R}^\omega$ with the box topology is **not locally path connected**.*

Proof. Let U be a basic open neighborhood of 0 by

$$U = \prod_{n \in \mathbb{N}} (-1/n, 1/n).$$

Pick a point $x = (x_1, x_2, \ldots) \in \mathbb{R}^\omega$ by setting

$$x_n = \frac{1}{2n}.$$

Since $x_n \in (-1/n, 1/n)$ for all $n \in \mathbb{N}$ we have $x \in U$.

Suppose a path $f : [0, 1] \to U$ with $f(0) = 0, f(1) = x$ exists.

Then

$$f_n = \pi_n \circ f : [0, 1] \to (-1/n, 1/n)$$

is a continuous real valued function with $f_n(0) = 0$ and $f_n(1) = 1/2n$.

By the intermediate value theorem, each f_n attains all values between 0 and $1/2n$.

In particular, there exists $t_n \in [0, 1]$ so

$$f_n(t_n) = \frac{1}{4n}.$$

We get a sequence $\{t_n\}_n \subseteq [0, 1]$, a compact metric space. Let $t^* \in [0, 1]$ be the limit of some convergent subsequence. Assume without loss of generality that

$$\lim t_n = t^*.$$

Let $f(t^*) = (y_1, y_2, \ldots) \in U$. Take

$$V = \prod_{n \in \mathbb{N}} (y_n - \varepsilon_n, y_n + \varepsilon_n)$$

where

$$\varepsilon_n = \frac{1}{2}\left|\frac{1}{4n} - y_n\right| < |f_n(t_n) - y_n|.$$

If $y_n = 1/4n$ set $\varepsilon_n = 1/8n$.

Then

$$\left|\frac{1}{4n} - y_n\right| \geq \varepsilon_n > 0.$$

Since V is open, there exists $\delta > 0$ so

$$|t - t^*| < \delta \Rightarrow f(t) \in V.$$

Since $\lim t_n = t^*$ choose N large enough so

$$|t_n - t^*| < \delta,\ n \geq N.$$

Then $f(t_n) \in V$.

Now, for all $n \geq N$ we have

$$\pi_n(f(t_n)) \in (y_n - \varepsilon_n, y_n + \varepsilon_n)$$

meaning

$$|f_n(t_n) - y_n| < \varepsilon_n.$$

Substituting we get

$$\left|\frac{1}{4n} - y_n\right| < \varepsilon_n,$$

but this contradicts

$$\left|\frac{1}{4n} - y_n\right| \geq \varepsilon_n.$$

■